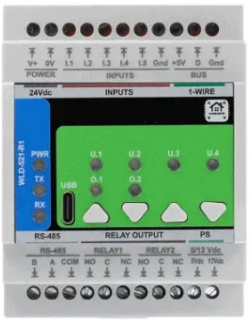
# Datasheet

## WLD-521-R1 Water Leak Detection Module




Part No.: WLD-521-R1

Manufacturer: ISYSTEMS AUTOMATION S.R.L.

Hardware Version: V1.0

Architecture: Modular (MCU Board + Field Board + Relay Extension)

Controller: RP2350A

## Functional Overview

The WLD-521-R1 is a modular water leak detection and water metering + I/O module for monitoring and automation. It provides 5 digital inputs, 2 SPDT relay outputs, 1-Wire sensor support, and optional 4 buttons / 4 user LEDs for local control and indication. The module connects to a PLC, MicroPLC, MiniPLC, SCADA, or other Modbus master as a Modbus RTU slave over RS-485 and is configured via USB-C Web Serial (WebConfig).

| Board | Function |
|---|---|
| MCU Board | RP2350A controller, RS-485 communication, USB-C configuration interface, power conversion |
| Field Board | 5 isolated digital inputs, isolated sensor power rails, 1-Wire interface |
| Relay Extension | 2 × SPDT relay contact outputs |

The module operates as a Modbus RTU slave over RS-485 and supports configuration and firmware update via USB-C.

## Technical Specifications

| Specification | Details |
|---|---|
| Microcontroller | RP2350A dual-core microcontroller |
| Storage | External QSPI Flash (W25Q32JV, 32 Mbit) |
| Power Input | 24 V DC nominal |
| Input Protection | 1 A fuse, reverse polarity diode, TVS surge suppression, EMI filtering |
| Main Logic Supply | Buck regulator 24 V → 5 V, 3.3 V LDO regulator |
| File system | LittleFS persistent configuration storage |
| Document revision | DS-WLD-521-R1 Rev. B · 2026-09 · Hardware R1 (V1.0) |
| Digital Inputs | 5 × isolated 5 V DC discrete inputs |
| Max Input Voltage | 5 V DC continuous |
| Relay outputs | 2 × SPDT (NO / C / NC); 3 A @ 250 VAC per contact (module limit) |
| Isolated +12 V Output | 12 V isolated rail, 2 W; max 167 mA theoretical ( ≤ 150 mA recommended) |
| Isolated +5 V Output | 5 V isolated rail, 1 W; max 200 mA theoretical ( ≤ 150 mA recommended) |
| 1-Wire | 1 × protected 1-Wire interface (+5 V, DATA, GND) |
| User Interface | 4 buttons; 9 LEDs (Power, 4 user, RX, TX, Relay1, Relay2) |
| RS-485 | half-duplex Modbus RTU, not galvanically isolated |
| USB | USB-C, 5 V logic, ESD protected |
| Typical Power Consumption | 3 W typical / 5 W maximum (both relays energised, isolated sensor supplies fully loaded) |
| Modbus defaults | Address 3, 19200 baud, 8N1 |

*Relay outputs are **not internally fused**. External overcurrent protection must be provided in accordance with applicable standards and installation requirements. Use appropriately rated fuses or circuit breakers and consider external contactors for loads exceeding 3 A or for inductive and high inrush loads.

Typical module power consumption: 3 W typical / 5 W maximum (both relays energised, isolated sensor supplies fully loaded).

### Top Terminal Block (1–11)

Front view – terminals numbered from left to right.

| Position | Marking | Signal | Description |
|---|---|---|---|
| 1 | V+ | POWER | 24 V DC input |
| 2 | 0V | POWER | 24 V DC return |

| | | | |
|---|---|---|---|
| 3 | I.1 | INPUTS | Input 1 |
| 4 | I.2 | INPUTS | Input 2 |
| 5 | I.3 | INPUTS | Input 3 |
| 6 | I.4 | INPUTS | Input 4 |
| 7 | I.5 | INPUTS | Input 5 |
| 8 | Gnd | INPUTS | Isolated input return (GND_ISO) |
| 9 | +5V | BUS | 1-Wire supply (logic domain, not isolated) |
| 10 | D | BUS | 1-Wire data |
| 11 | Gnd | BUS | 1-Wire ground (logic) |

### Bottom Terminal Block (1–11)

Front view – terminals numbered from left to right.

| Position | Marking | Signal | Description |
|---|---|---|---|
| 1 | B | RS485 | RS-485 data - |
| 2 | A | RS485 | RS-485 data + |
| 3 | COM | RS485 | RS-485 signal reference |
| 4 | NO | RELAY1 | Relay 1 normally open |
| 5 | C | RELAY1 | Relay 1 common |
| 6 | NC | RELAY1 | Relay 1 normally closed |
| 7 | NO | RELAY2 | Relay 2 normally open |
| 8 | C | RELAY2 | Relay 2 common |
| 9 | NC | RELAY2 | Relay 2 normally closed |
| 10 | 5Vdc | PS | Isolated sensor rail +5 V |
| 11 | 12Vdc | PS | Isolated sensor rail +12 V |

## Installation, Environmental & Mechanical

| Specification | Details |
|---|---|
| Terminal Type | Pluggable screw terminal blocks, 5.08 mm pitch |
| Wire Cross Section | 0.2–2.5 $mm^2$ (AWG 24–12) |
| Tightening Torque | 0.4–0.6 Nm |

| Specification | Details |
|---|---|
| Operating temperature | 0 °C to +40 °C |
| Storage temperature | −10 °C to +55 °C |
| Relative humidity | 0–90 % RH, non-condensing |

| | |
|---|---|
| Ingress protection | IP20 (inside cabinet only) |
| Installation | Indoor control cabinet only; not for outdoor or exposed installation |
| Maximum altitude | 2000 m |
| Pollution degree | 2 |

| Specification | Details |
|---|---|
| Dimensions | 70 × 90.6 × 67.3 mm (L × W × H) |
| DIN width | 4 modules ( ≈ 70 mm) |
| Mounting | 35 mm DIN rail |
| Enclosure | PC/ABS industrial enclosure |
| Terminal type | Pluggable screw terminal blocks, 5.08 mm pitch |
| Wire cross-section | 0.2–2.5 mm² (AWG 24–12) |
| Tightening torque | 0.4–0.6 Nm |
| Net weight | 200 g |
| Gross weight | 320 g |
| Pack size | 140 × 125 × 94 mm (L × W × H) |
| Mechanical drawing | Mechanical drawing: front and side view, dimensions in mm |
| Notes | All dimensions shown in millimeters unless stated otherwise |

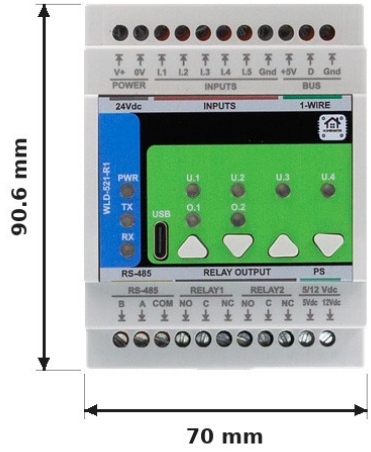
Install only inside a control cabinet with ventilation; the cabinet must include a protective front plate covering all module connection terminals and a closing protective door; not for outdoor or exposed installation.

All wiring terminals must be protected against accidental contact by an insulating front plate, wiring duct, or terminal cover. Exposed live terminals are not permitted.

# WLD-521-R1

All dimensions in mm

Front view




Side view



## Cable Recommendations & Shield Grounding

Use shielded, twisted constructions and bond shields correctly to reduce EMI and ground-loop issues.

### General Routing Rules

- Route low-level signal cables separately from power wiring.
- If crossing power cables is unavoidable, cross at 90°.
- Keep cable runs as short as practical.
- Avoid parallel runs with high-current conductors.

### RS-485 Cable

- Twisted pair recommended (120 Ω characteristic impedance).
- Shielded cable recommended in industrial environments.
- Use one twisted pair for A/B.
- Run COM to every node. Required, not optional: the RS-485 port is not galvanically isolated.

### Shield Grounding

- Bond cable shield(s) to cabinet PE/EMC ground at one end (PLC side recommended).
- Do not connect shields directly to signal terminals (A/B/COM).

## Compliance & Certifications

The WLD-521-R1 module is CE marked and designed to comply with applicable European Union directives. The manufacturer maintains technical documentation and a signed EU Declaration of Conformity (DoC).

EMC 2014/30/EU · LVD 2014/35/EU · RoHS 2011/65/EU

Standards: EN 61000-6-1 · EN 61000-6-3 · EN 62368-1 · EN IEC 63000

## Safety Notice

The power input and signal terminals are SELV circuits. Relay output terminals may switch hazardous voltage up to 250 VAC.

Adequate insulation and separation must be ensured during installation.Installation by qualified personnel only.

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